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1 Theory:

The line following robot is a simple robot that follows a specific path indicated by a line (usually a black line on a light-coloured surface) having some width. The robot is structured with the chassis of an omni directional robot in mind. Primary components that ensure the proper operation of the robot are infra-red sensors and a micro-controller. The paper discusses the circuit diagram, and the chassis design in the upcoming sections.

1.1 Working Of The IR Sensor:

The IR transmitter continuously transmits the IR rays. When IR transmitter is on the black surface these rays were absorbed by the surface and when it is on white surface these rays were reflected. The IR receiver has maximum resistance when no IR rays are received and voltage from VCC flows through the resistor. At the output pin, voltage is approximately 5V.

As the intensity IR rays received by the receiver increases, resistance value decreases and reverse break down occurs. Thus, voltage through the resistor is grounded. So, at the output pin, it will produce 0V.

2 Components In The Circuit:

- Bluepill Microcontroller – STM32C8.
- Two Encoder Motors.
- Two Story Based Chassis.
- IR sensors.
- Motor Drivers.

3 3D Model And Chassis Dimensions:

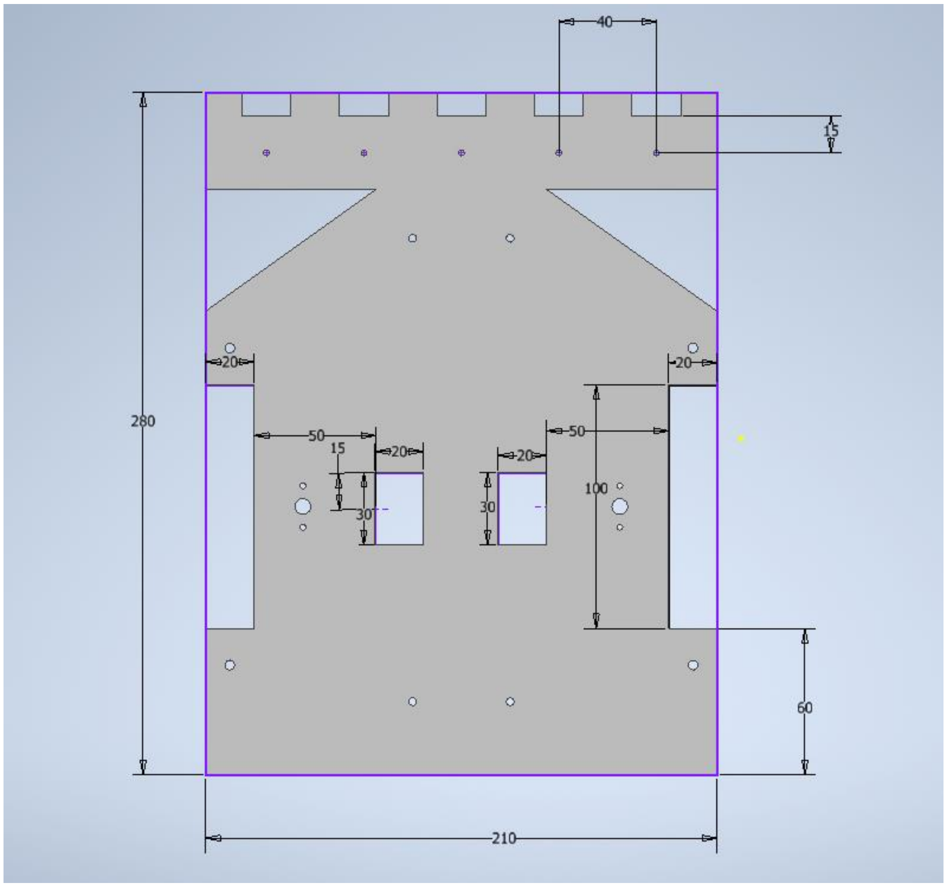

Figure: Base of the robot.

3.1 Essential Dimensions

Chassis Body	Length = 280 mm, Width = 210 mm
Wheel Cut Out	Length = 100 mm, Width = 20 mm
Motor Connections Allowance Cutout	Length = 30 mm, Width = 20mm
Screw Hole	Radius = 5mm

4 Circuit Layout And *STM Pin Outs:*

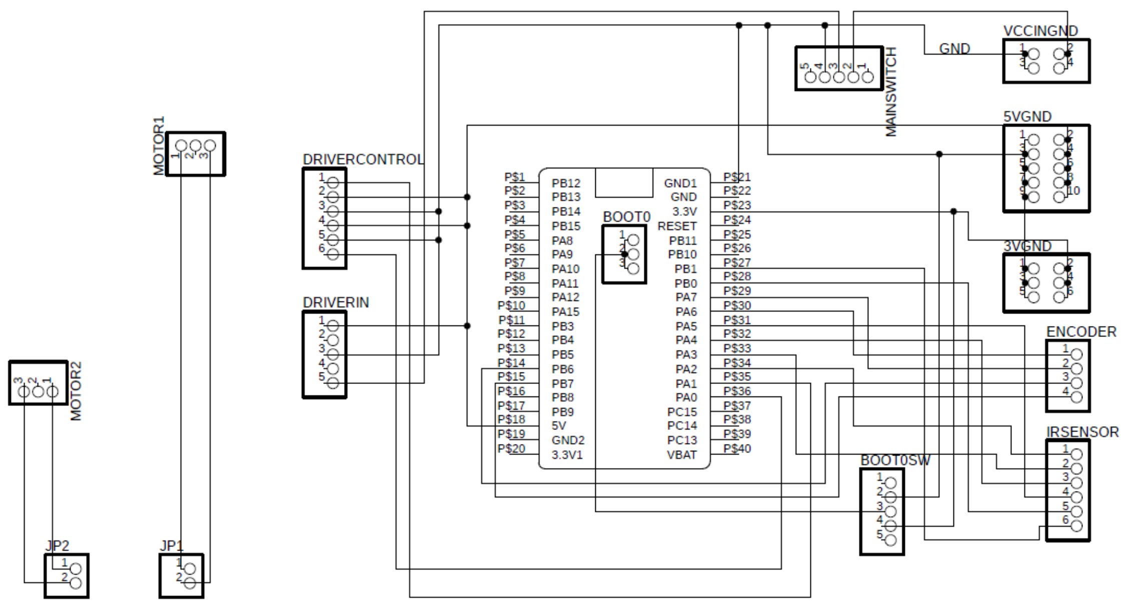

Figure: Circuit Diagram

4.1 Essential STM Pin Outs:

Pinouts	**Functions**
A9 A10	Program
A6 A7	Input From Encoder 1
B6 B7	Input From Encoder 2
A2	Input From Left IR Sensor
A5	Input From Center IR Sensor
B0	Input From Right IR Sensor
A0	PWM Generation For Right Motor
A1	PWM Generation For Left Motor

5 Code:

```
while (1)
 {
		If (HAL_GPIO_ReadPin (GPIOA, GPIO_PIN_5))
		{
			duty_r = 10000;
			duty_l = 10000;

				if(HAL_GPIO_ReadPin (GPIOA, GPIO_PIN_2)) //sensor left
				{
					duty_r = duty_r+2000;
					duty_l = duty_l-6000;
				}
				if(HAL_GPIO_ReadPin (GPIOB, GPIO_PIN_0)) // sensor right
				{
					duty_r = duty_r-6000;
					duty_l = duty_l+2000;
				}
		}// 1 if HIGH, 0 if low****sensor1
		else if(HAL_GPIO_ReadPin (GPIOA, GPIO_PIN_2))//sensor left
		{
			duty_r = 8000;
			duty_l = 15000;
		}
		else if(HAL_GPIO_ReadPin (GPIOB, GPIO_PIN_0)) // sensor right
		{
			duty_r = 15000;
			duty_l = 8000;
		}
		else
		{
			duty_r = 0;
			duty_l = 0;
		}
  /* USER CODE END WHILE */
		TIM2->CCR1= duty_r;
		TIM2->CCR2= duty_l;
  /* USER CODE BEGIN 3 */
 }
 /* USER CODE END 3 */
}
```

The above code is embedded in the main code module of Kiel. This code snippet excludes all the initializations. Rather, it includes the code of PWM control which is derived using a simplified rule base on the following parameters.

- If center sensor is sensing the line while the other two sensors don’t, the velocity of both wheels are the same.
- If left sensor is sensing the line while the other two sensors don’t, the velocity of right wheel is increased and that of left wheel is decreased enabling the robot to move left.
- If right sensor is sensing the line while the other two sensors don’t, the velocity of left wheel is increased and that of right wheel is decreased enabling the robot to move right.
- Line sensing by both center and right sensor indicates a sharp right turn. Therefore, to accommodate the sudden turn the speed of left wheel is increased to considerable amount, while the velocity of right wheel is tuned very low.
- Line sensing by both center and left sensor indicates a sharp left turn. Therefore, to accommodate the sudden turn the speed of right wheel is increased to considerable amount, while the velocity of left wheel is tuned very low.
- Both wheels stop when all three sensors are high or low.

6 Discussion And Improvements:

Laser Cutting technology was used to imprint our design dimensions on raw acrylic sheets. These made the base and top story of our chassis with each fastened together using bolts. To aid in movement and balance the robot we also incorporated two castor wheels. To ensure the operation of the wheel and prevent inertia and static friction from hindering the commencement of wheel motion, we tried and tested several PWM values and selected a fail-proof lowest PWM that ensured the movement of wheel.

Currently, the robot’s wheel is controlled by directly controlling the PWM. Another way to do that is using speed control and alter PWM according to the desired speed. Moreover, increasing the number of sensors, and intaking analog input from sensors will provide better control and reduce the jitteriness in the movement.